

# Terentiy Gatsukov

## Senior C# / .NET Developer | Slot Games

Minsk, Belarus · Remote; open to relocation for a strong offer

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### PROFILE

**5+ years of experience, including about 3 years developing slots for myKONAMI.** Independently built the backend for **approximately 15 games from scratch to production**, from control-file parsing to deployment. Shared C# / F# library used in **20+ titles**; simulations of **1 billion+ spins** and a **6x** calculation speedup.

### CORE SKILLS

**Game math & simulation:** RTP, volatility, hit rates, cycles, discrepancy analysis, API simulations.  
**Algorithms:** probability monads, trampolines, unbiased RNG, alias method, segment trees.  
**Engineering:** C#, F#, TypeScript, .NET, memory optimisation, Roslyn, concurrency correctness.

### EXPERIENCE

#### .NET Developer · Custom Games Studio

November 2023 - July 2026 · Minsk

##### Slots for myKONAMI · Game engine and backend platform (5M+ users)

- **Designed approximately 90% of a shared library from scratch for 20+ titles:** monads, trampolines, samplers and simulation engine. Defined architectures for new mechanics and reviewed code in a team of 4 backend developers.
- **Implemented Bind/Map and Select/SelectMany in C# and composition in F#:** sampling and exhaustive outcome enumeration with exact probability calculation. Built trampolines to execute nested bonuses, respins and cascades without stack growth.
- **Implemented rejection sampling without modulo bias, the alias method and segment trees:** dynamic weights, sampling without replacement, distribution and boundary-value checks.
- **Built control-file processing:** XML / JSON parsing and conversion, validation of weights, reel strips, paytables and cross-configuration references; .NET class generation.
- **Built local and API-based simulations and automated regression checks:** state machines, transitions, payouts, event persistence and Kafka publishing; reconciled results with mathematicians.
- **Scaled simulations of 1 billion+ spins across 2-4 machines:** separate RNG streams, checkpoints, resumable runs and worker-statistics aggregation; replay using seeds, RNG state or recorded random values.
- **Calculated expected values, variance, standard deviation, standard error and confidence intervals,** run counts for target precision, base/bonus RTP, hit rates, cycles, symbol averages, payout distributions and nested-feature metrics.
- **Implemented Couchbase state persistence and recovery** after restarts and reconnects; verified idempotency and duplicate-operation protection. Nexus was the source of truth for balances.
- **Reduced calculation time from 3 hours to 30 minutes:** Span<T>, ArrayPool<T>, readonly structs, fewer allocations and lower GC pressure. A Roslyn Source Generator improved serialisation speed by ~35% across 10+ components.
- **Ensured correct bet processing at 5,000+ RPS:** Redis and conditional SQL updates prevented double charges. Moved events to Kafka, reducing game-operation p99 latency to <50 ms.

### GAMES & IMPLEMENTED MECHANICS

**Titles:** Unwooly Riches, Diamond Rider, Lucky Drums, Dragon's Law Fortune Pots, Mayan Chief, Great Moai and others. Additional portfolio links on page 2.

**Mechanics:** Super Respin, Hold & Spin, Wheel Bonus, jackpots, progressives, free spins; expanding / double reels, multipliers, sticky / mystery / dropping wilds; collection mechanics; Cascading Avalanche, Megaways, Cluster, Nudge.

## Terentiy Gatsukov

### SELECTED GAMES · myKONAMI

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My role: full game backend. Links open official Konami Gaming video demonstrations.

[Unwooly Super Riches](#)

[K Pow! Pig](#)

[Charms Full Link](#)

[Lucky Honeycomb Fortune](#)

[Dragon's Law Fortune](#)

[Prize Strike](#)

[Lucky Drums](#)

### PREVIOUS EXPERIENCE

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#### C# / .NET Developer · Solvintech

July 2022 - October 2023

[MPsklad](#) · [Marketplace SaaS](#) · [Ozon](#), [Wildberries](#), [Yandex Market](#), [MoySklad](#) · 50k+ orders/day

- **Ensured order and event consistency** with Transactional Outbox / Inbox in PostgreSQL and Kafka. Eliminated event loss and status inconsistencies during failures and redelivery.
- **Prevented inventory overbooking under concurrent orders:** distributed Redis coordination and Kafka ordering by SKU prevented duplicate deductions and discrepancies.
- **Reduced order processing time to <120 ms.** Separated writes and reads with CQRS (PostgreSQL / MongoDB), moving marketplace integrations to asynchronous Kafka and Hangfire processing.

**Stack:** C#, .NET, ASP.NET Core, EF Core, PostgreSQL, MongoDB, Redis, Kafka, Hangfire, Docker, React, TypeScript, xUnit, Testcontainers.

#### C# / .NET Developer · Elgrow

January 2021 - June 2022 · Remote

[Domodedovo Airport smart parking](#) · 1,000+ spaces · 10,000+ users

- **Designed and built a booking system** with ASP.NET Core, RabbitMQ and SignalR. Achieved API p99 latency of <200 ms under peak load.
- **Isolated services from equipment failures:** moved barrier-controller communication to asynchronous RabbitMQ messaging; integrated ALPR cameras and access-control systems.
- **Built mobile REST APIs with payments and push notifications.** Ensured payment idempotency and verified integration scenarios with xUnit / Docker.

**Stack:** C#, .NET 5, ASP.NET Core, PostgreSQL, EF Core, Redis, RabbitMQ / MassTransit, SignalR, Polly, Docker, xUnit, Testcontainers, Firebase.

### EDUCATION & LANGUAGES

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#### Belarusian State University of Informatics and Radioelectronics

Faculty of Computer-Aided Design, Minsk · Engineering degree, 2023.

Specialisation: Design and Production of Software-Controlled Electronic Devices.

**Russian:** native. **English:** B1; reading technical documentation.

**Citizenship:** Belarus. **Work authorisation:** Armenia, Belarus, Georgia, Kazakhstan, Russia, Uzbekistan.